

CSE 5410

Lab 3

Gaven Flores

008493310

1. Lab

In this lab the experiment uses a photo-interrupter, a photo interrupter consists of a transmitter (an IR LED) and receiver (a phototransistor) that face each other the transmitter emits a light into the receiver. When light hits the phototransistor, it conducts and SIG goes LOW. When the light is blocked the phototransistor stops conducting and SIG becomes HIGH. The hardware is wired from GND to GND , Vcc to 5v and SIG to pin 8. The goal of this lab is to invert the original behavior, so the LED turns on when blocked and off when unblocked.

2. Experiment

In the code to invert the original behavior I changed `const int photoPin = 7` to 8 and i flipped the order of the `digitalWrite` statements in the if block. Flipping the `digitalWrite` statements inverts the behavior because the original code set the LED LOW (off) when the sensor read HIGH, so the swap allows to set the LED HIGH (on) when blocked instead. Originally when i tried to test the circuit I ran into a problem but then when nothing was happening when placing the paper between transmitter and receiver, I re looked at my wiring and realized I had SIG plugged into 7 and not 8.

3. Conclusion

This lab taught me that even with the simplest changes in both software and hardware that things can go wrong like having SIG plugged into the wrong pin initially. Also, it really showed me the importance of having at least a basic grasp on what is going on between the steps where you write the software, compile it to the board and during execution. For example, if you didn't understand the original code then it would make it much more difficult to figure out what you need to swap to invert the initial logic.

4. Attachments

Modified code

```
    const int photoPin = 8;
const int ledPin = 13;

void setup()
{
    pinMode(photoPin, INPUT);
    pinMode(ledPin, OUTPUT);
}

void loop()
{
    boolean Value=digitalRead(photoPin);

    if(Value==HIGH)
    {
        digitalWrite(ledPin, HIGH);
    }
    else
    {
        digitalWrite(ledPin, LOW);
    }
}
```

Picture of circuit

